Sharif University of Technology

Department of Mechanical Engineering

Renewable Energy Systems Engineering – Phase II

Design and Feasibility Study of an Integrated Polygeneration System for a Green Data Center

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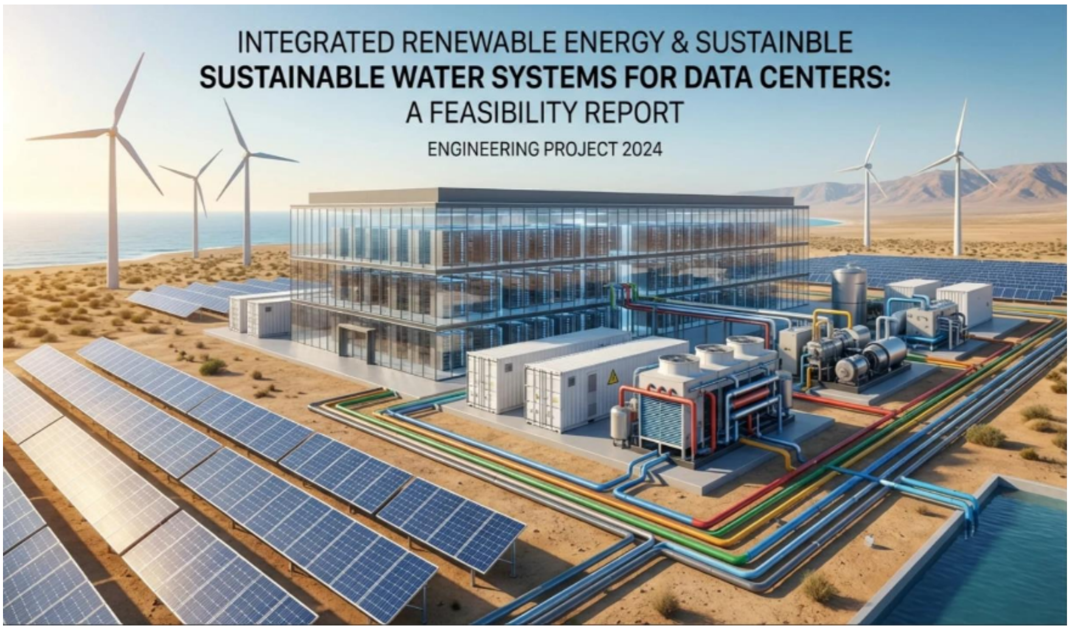
INTEGRATED RENEWABLE ENERGY & SUSTAINABLE WATER SYSTEMS FOR DATA CENTERS



Conceptual Infrastructure Layout of the Chabahar Polygeneration Green Data Center

Course Instructor: Dr. Bijarchi

Project Authors: **Sina Zerehpoosh (Student ID: 401170829)**
Tina Movaseghi (Student ID: 401170961)
Shayan Abedi (Student ID: 401170886)

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Chapter 1

Introduction and Problem Statement

1.1 Background and Global Significance

Data centers serve as the fundamental backbone of the modern digital economy, representing significant consumers of global electrical energy. Worldwide statistics show that data centers currently consume between 1.3% and 2.0% of total global electricity generation. Driven by the rapid proliferation of:

- Cloud Computing infrastructures
- Artificial Intelligence (AI) and High-Performance Compute clusters
- Internet of Things (IoT) ecosystems

projections indicate that by 2030, data center electricity consumption will approach 3.0% of the world's total electricity supply.

1.2 Core Engineering Challenges

Guaranteeing 24/7 continuous operation while maintaining thermal reliability poses two interconnected engineering hurdles: continuous uninterrupted electrical supply and high-density heat dissipation. Conventional facilities present several environmental and operational liabilities:

1. Substantial quantities of valuable thermal heat are dumped directly into the ambient environment via evaporative cooling towers.
2. Conventional evaporative systems consume unsustainable volumes of freshwater.
3. Massive localized electrical demands place severe peak stress on utility grids.
4. Heavy reliance on fossil-fuel-based grid electricity generates excessive carbon dioxide (CO_2) emissions.

1.3 Project Scope and Objectives

This project develops an integrated polygeneration facility designed for a modern high-density data center, satisfying the following objectives:

- 100% continuous power supply derived from hybrid solar photovoltaic and wind energy resources.
- Low-grade waste heat recovery from liquid-cooled compute racks to drive secondary electricity generation.
- Thermally driven cooling systems satisfying the entire data center heat dissipation demand.
- Autonomous on-site seawater desalination, eliminating external municipal freshwater demands.
- Absolute minimization of grid dependency and fossil fuel consumption.

Chapter 2

Geographic and Climatic Analysis of Chabahar

The chosen deployment site is located in the coastal city of Chabahar, situated within Sistan and Baluchestan Province along the Makran coastline of the Sea of Oman:

- **Latitude:** 25.29° N
- **Longitude:** 60.64° E

2.1 Solar Resource Assessment

Chabahar features superior solar potential, offering more than 300 cloudless sunny days per year and an annual average Global Horizontal Irradiance (GHI) of 5.87 kWh/m^2/day. Monthly irradiation data and ambient temperatures are presented in Table 2.1.

Table 2.1: Monthly solar irradiation and temperature data for Chabahar.

Month	**GHI** [kWh/m^2/day]	**DNI** [kWh/m^2/day]	**Mean Temp.** [°C]
Farvardin (Apr)	6.2	5.8	26
Ordibehesht (May)	7.1	6.5	30
Khordad (Jun)	7.3	6.2	33
Tir (Jul)	6.5	5.0	34
Mordad (Aug)	6.2	4.8	33
Shahrivar (Sep)	6.4	5.5	32
Mehr (Oct)	5.9	5.6	30
Aban (Nov)	5.2	5.0	27
Azar (Dec)	4.5	4.2	24
Dey (Jan)	4.3	4.0	22
Bahman (Feb)	5.0	4.8	22
Esfand (Mar)	5.8	5.5	24
Annual Mean	**5.87**	**5.24**	**28**

2.2 Wind Resource Assessment

Coastal sea-breeze circulations combined with regular summer and winter monsoon wind patterns establish a favorable wind generation environment, as summarized in Table 2.2.

Table 2.2: Wind resource characteristics at 100 m hub height.

Resource Parameter	**Design Value**
Mean Wind Speed (100 m Hub Height)	6.5 m/s
Wind Power Density	280 W/m^2
Dominant Wind Direction	South-West (Monsoon)
Peak Wind Season	June to September (Khordad to Shahrivar)
Estimated Annual Capacity Factor (CF)	30% – 35%

Chapter 3

Data Center Baseline Design Specifications

The baseline IT workload and power distribution boundaries governing the sizing of the power facility are summarized in Table 3.1.

Table 3.1: Data center electrical capacity and technical requirements.

Design Parameter	**Design Value**	**Unit**
Total IT Electrical Load	2.0	MW
Total Server Rack Count	100	Units
Power Rating per Rack	20	kW/rack
Facility Distribution Voltage	400	VAC
Frequency	50	Hz
Reliability Standard	Continuous 24/7 (Tier III Target)	—

Chapter 4

Solar Photovoltaic Power Plant Design

4.1 Resource Advantages and Challenges

- **Key Advantages:** Intense horizontal irradiance, direct coincidence with peak daytime power demand, negative correlation with daytime wind minimums, established commercial maturity, minimal maintenance overhead, and a 25-year service life.
- **Operational Challenges:** Dust accumulation necessitating regular automated cleaning, elevated cell temperatures inducing thermal derating, and absence of output during nighttime hours.

4.2 Capacity Sizing Formulations

Table 4.1: Design parameters for solar PV sizing.

Parameter	**Value**	**Unit**
Baseline Facility Load (P_{load})	$2,000$	kW
Solar Generation Share	50%	—
Peak Sun Hours (PSH)	5.2	h/day
System Performance Ratio (PR)	80%	—
Temperature Derating Factor (η_{temp})	0.92	—

The daily electrical energy demanded from the solar array is calculated as:

$$E_{\text{daily}} = \frac{P_{\text{load}} \times 24 \times 0.50}{\text{PR}} = \frac{2000 \times 24 \times 0.50}{0.80} = 30,000\,\text{kWh/day} \tag{4.1}$$

The minimum required DC installed peak power without safety margins is:

$$P_{\text{pv}} = \frac{E_{\text{daily}}}{\text{PSH} \times \eta_{\text{temp}}} = \frac{30000}{5.2 \times 0.92} \approx 6,270\,\text{kW}_{\text{p}} \tag{4.2}$$

Applying an engineering safety and battery-charging margin of 1.25:

$$P_{\text{pv,installed}} = 6,270 \times 1.25 \approx 8,000\,\text{kW}_\text{p} = 8.0\,\text{MW}_\text{p} \tag{4.3}$$

4.3 PV Module Technology Selection

Table 4.2: Commercial photovoltaic cell technology comparison.

Technology	**Efficiency**	**Advantages**	**Disadvantages**
Mono-PERC Bifacial	**$20-22\%$**	**High efficiency,** $+5-15\%$ **rear yield**	**Higher initial CapEx**
Poly-Si	$17-19\%$	Lower upfront capital cost	Lower energy density
Thin-Film	$12-14\%$	Favorable temp. coefficient	Large footprint required

Table 4.3: Technical specifications of the selected Mono-PERC Bifacial module.

Module Specification	**Value / Rating**
Cell Architecture	Mono-PERC Bifacial
Nominal Peak Output (W_p)	$545\,\text{W}_\text{p}$
Module Conversion Efficiency	21.3%
Dimensions	$2278 \times 1134 \times 35\,\text{mm}$
Unit Weight	28.6 kg
Temperature Coefficient (P_{max})	$-0.35\%/°\text{C}$
Power Retention Guarantee	25 Years (84.8% guaranteed output)
Target Manufacturers	LONGi Solar, JA Solar, Trina Solar

Total module quantity (N_{pv}) and land requirements are computed as follows:

$$N_{\text{pv}} = \frac{P_{\text{pv,installed}}}{P_{\text{rated}}} = \frac{8 \times 10^6\,\text{W}}{545\,\text{W}} \approx 14,700\,\text{Modules} \tag{4.4}$$

$$A_{\text{panel}} = 2.278 \times 1.134 = 2.583\,\text{m}^2 \implies A_{\text{net}} = 14,700 \times 2.583 = 37,970\,\text{m}^2 \tag{4.5}$$

For single-axis tracking mounts with a Ground Cover Ratio (GCR = 0.35):

$$A_{\text{land}} = \frac{A_{\text{net}}}{\text{GCR}} = \frac{37,970}{0.35} = 108,486\,\text{m}^2 \approx 10.7\,\text{ha} \tag{4.6}$$

Table 4.4: Executive summary of the solar PV installation.

System Parameter	**Value**
Installed DC Peak Capacity	$8.0\,\mathrm{MW_p}$
Total Module Count	$14,700$ Units
Tracking Mechanism	Single-Axis Horizontal Tracker
Central Inverters	$4 \times 2.5\,\mathrm{MW}$ Units
Land Footprint	$10.7\,\mathrm{ha}$
Anticipated Annual Generation	$\approx 15,000\,\mathrm{MWh/year}$
Capacity Factor	$\approx 21\%$

Chapter 5

Wind Power Plant Design

5.1 Synergy and Wind Characteristics

The local wind resource at Chabahar provides continuous complementary generation: coastal monsoons peak during summer months and sustain generation during evening and nighttime periods. The local Weibull parameters are summarized in Table 5.1.

Table 5.1: Weibull distribution parameters for Chabahar.

Parameter	**Value**
Weibull Shape Factor (k)	2.1
Weibull Scale Factor (c)	7.3 m/s
Annual Mean Wind Speed	6.5 m/s

5.2 Capacity Sizing and Turbine Selection

Table 5.2: Design baseline for wind farm sizing.

Parameter	**Value**	**Unit**
Continuous Data Center Load	$2,000$	kW
Target Wind Energy Share	50%	—
Target Capacity Factor (CF)	32%	—
Full-Load Operating Hours	$2,800$	h/year

The required theoretical rated capacity is:

$$P_{\text{wind}} = \frac{E_{\text{annual}}}{\text{CF} \times 8760} = \frac{8,760,000\,\text{kWh}}{0.32 \times 8760\,\text{h}} = 3,125\,\text{kW} \tag{5.1}$$

To align with standard commercial turbine sizes, installed capacity is set to 4.0 MW using two 2.0 MW units.

Table 5.3: Comparative evaluation of commercial utility-scale wind turbines.

Turbine Model	Rating	Rotor Diameter	IEC Class	Chabahar Suitability
Vestas V90-2.0	2.0 MW	90 m	IIA	Moderate
Vestas V110-2.0	**2.0** MW	**110** m	**IIIA**	**Optimal**
Siemens SG 2.6-114	2.6 MW	114 m	IIA	Good
Enercon E-92	2.35 MW	92 m	IIA	Moderate

Table 5.4: Detailed technical specifications of the Vestas V110-2.0MW wind turbine.

Technical Metric	Specification / Value
Rated Generator Power	2.0 MW
Rotor Diameter (D)	110 m
Rotor Swept Area	$9,503\,\mathrm{m}^2$
Hub Elevation	95 m
Cut-in Wind Speed	3.0 m/s
Rated Wind Speed	11.5 m/s
Cut-out Wind Speed	25.0 m/s
IEC Wind Class	Class IIIA (Low to Medium Wind Regimes)
Generator Topology	Doubly-Fed Induction Generator (DFIG)
Design Operational Life	25 Years

5.3 Micrositing, Footprint, and Annual Energy Yield

To mitigate wake interference losses:

- **Crosswind Spacing:** $4D = 4 \times 110 = 440\,\mathrm{m}$
- **Downwind Spacing:** $7D = 7 \times 110 = 770\,\mathrm{m}$

Dedicated buffer area per turbine:

$$A_{\text{turbine}} = 440 \times 770 = 338,800\,\mathrm{m}^2 = 33.88\,\mathrm{ha} \tag{5.2}$$

$$A_{\text{windfarm}} = 2 \times 33.88\,\mathrm{ha} = 67.76\,\mathrm{ha} \approx 68\,\mathrm{ha} \tag{5.3}$$

Note: Only the foundation bases occupy ground area ($< 1\%$); intermediate buffer land remains usable.

Annual Energy Production (AEP) evaluation:

$$\mathrm{AEP} = P_{\text{rated}} \times N_{\text{turbines}} \times \mathrm{CF} \times 8760 = 2000 \times 2 \times 0.32 \times 8760 = 11,212\,\mathrm{MWh/year} \tag{5.4}$$

Table 5.5: Simulated monthly wind generation profile.

Month	**Capacity Factor (CF)** [%]	**Monthly Generation** [MWh]
Farvardin (Apr)	28	806
Ordibehesht (May)	30	893
Khordad (Jun)	38	1,095
Tir (Jul)	40	1,190
Mordad (Aug)	42	1,249
Shahrivar (Sep)	38	1,095
Mehr (Oct)	32	952
Aban (Nov)	28	806
Azar (Dec)	26	749
Dey (Jan)	24	714
Bahman (Feb)	26	749
Esfand (Mar)	28	806
Total / Average	**32%**	**11,104**

Chapter 6

Battery Energy Storage System (BESS) Design

6.1 System Rationale and Critical Scenarios

Energy storage is required to bridge the intermittency of renewable generation and provide uninterrupted power for the IT infrastructure. Three critical operational design cases are considered:

Table 6.1: Critical operational sizing scenarios for the BESS.

Scenario Description	Duration	Required Power	Required Energy
Scenario 1: Calm Night (Zero PV & Wind)	6.0 h	2,300 kW	13,800 kWh
Scenario 2: Overcast Day with Calm Winds	Full Day	Partial Deficit	$\approx 10,000$ kWh
Scenario 3: Peak Load during Transient Drop	4.0 h	3,000 kW	12,000 kWh

6.2 Storage Capacity Sizing

Table 6.2: Design parameters for BESS sizing.

Parameter	Value	Engineering Basis
Backup Autonomy Window	6.0 h	Maximum calm night duration
Continuous Load (IT + Auxiliaries)	2,300 kW	$2000\,\text{kW}_{\text{IT}} + 300\,\text{kW}_{\text{aux}}$
Depth of Discharge (DoD)	80%	Preserves cell cycle life
Round-Trip AC-AC Efficiency (η_{RT})	92%	High-efficiency bidirectional inverter
Oversizing Safety Margin	1.15	Buffers against cell aging and fade

Mathematical capacity derivation:

$$E_{\text{required}} = P_{\text{load}} \times t_{\text{backup}} = 2300\,\text{kW} \times 6\,\text{h} = 13,800\,\text{kWh} \tag{6.1}$$

$$E_{\text{nominal}} = \frac{E_{\text{required}}}{\text{DoD} \times \eta_{\text{RT}}} = \frac{13800}{0.80 \times 0.92} \approx 18,750\,\text{kWh} \tag{6.2}$$

$$E_{\text{BESS}} = 18,750 \times 1.15 \approx 21,563\,\text{kWh} \implies E_{\text{installed}} = 28.0\,\text{MWh} \tag{6.3}$$

Capacity is specified at 28.0 MWh *to cover cell degradation, multi-day cloud cover, and Tier III redundancy.*

6.3 Cell Chemistry Selection and Container Configuration

Table 6.3: Comparative evaluation of commercial utility-scale battery chemistries.

Chemistry	Energy Density	Cycle Life	Thermal Safety	Suitability
LFP ($LiFePO_4$)	**120** Wh/kg	**6,000+**	**Superior**	**Optimal**
NMC	180 Wh/kg	3,000	Moderate	Thermal runaway risk
LTO	80 Wh/kg	15,000+	Superior	High capital expense
Flow Battery	25 Wh/L	20,000+	Excellent	Large land footprint
Lead-Acid	35 Wh/kg	1,500	Fair	Obsolete

Table 6.4: Executive specifications of the containerized BESS installation.

Parameter	Value / Specification
Installed Nominal Energy	28.0 MWh
Usable Capacity (80% DoD)	22.4 MWh
Battery Chemistry	Lithium Iron Phosphate (LFP)
Inverter Rated Power	7.0 MW
Backup Duration	4.0 – 6.0 h
Enclosure Architecture	20 Standard 40 ft High-Cube ISO Containers
Container Rating	1.4 MWh per container (≈ 28 tons/unit)
AC-to-AC Round-Trip Efficiency	> 92%
Installation Land Footprint	0.15 ha
Operating Lifespan	15 Years

Chapter 7

Organic Rankine Cycle (ORC) Waste Heat Recovery

7.1 System Rationale

Direct-to-chip liquid cooling circuits extract heat from server processors at temperatures between 55°C and 60°C. Because this low temperature is unsuitable for traditional steam cycles, an Organic Rankine Cycle (ORC) utilizing a low-boiling organic working fluid is used to recover electrical energy.

7.2 Working Fluid Evaluation

Table 7.1: Thermophysical properties of candidate ORC working fluids.

Fluid	T_{boil} [°C]	GWP	ODP	Safety Class	Suitability
R245fa	**15.3**	**858**	**0**	**A1 (Non-toxic/flam.)**	**Optimal**
R1233zd(E)	19.0	< 1	0	A1	Good
Isopentane	27.8	< 10	0	A3 (Flammable)	Safety Hazard

7.3 Performance Modeling and Output Ratings

Table 7.2: Technical specifications of the low-temperature ORC recovery unit.

Parameter	Design Value
Qualified Manufacturers	Turboden, Ormat, Againity
Net Electrical Output	$50\,\text{kW}_\text{e}$
Working Fluid	R245fa
Heat Source Temperature	55°C (Hot water from compute racks)
Condensation Temperature	28°C (Seawater cooled)
Thermal Conversion Efficiency	7.6%
Turbine Expander Type	Radial Inflow Impeller
Skid Dimensions	$3.0 \times 2.0 \times 2.5\,\text{m}$

Thermodynamic conversion balance:

$$\eta_{\text{th,ideal}} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_{\text{in}}} = \frac{50}{500} = 10.0\% \implies \eta_{\text{actual}} \approx 7.6\% \tag{7.1}$$

$$P_{\text{ORC}} = 50\,\text{kW}_{\text{e}} \implies E_{\text{ORC,annual}} = 50\,\text{kW} \times 8000\,\text{h} = 400\,\text{MWh/year} \tag{7.2}$$

This recovered energy is fed directly into internal data center AC distribution buses, reducing grid draw, buffering BESS discharge, and lowering facility Power Usage Effectiveness (PUE).

Chapter 8

Absorption Chilling System

8.1 System Rationale

Mechanical vapor compression chillers consume significant electrical power ($0.6 - 0.9\,\mathrm{kW_e/ton}$), increasing overall PUE. In contrast, single-effect absorption chillers consume minimal electrical power, converting thermal energy from liquid cooling circuits directly into chilled water.

Table 8.1: Design parameters of the single-effect absorption chiller.

Parameter	**Specification / Value**
Chiller Architecture	Single-Effect Absorption
Refrigerant–Absorbent Pair	Water–Lithium Bromide ($H_2O{-}LiBr$)
Cooling Capacity ($\dot{Q}_{\mathrm{cooling}}$)	$900\,\mathrm{kW_c}$
Thermal Driving Input ($\dot{Q}_{\mathrm{in}}$)	$1,200\,\mathrm{kW_{th}}$
Coefficient of Performance (COP)	0.75
Hot Water Inlet Temperature	$70°\mathrm{C} - 85°\mathrm{C}$
Chilled Water Supply Temperature	$7.0°\mathrm{C}$
Parasitic Pumping Electrical Load	$18.0\,\mathrm{kW_e}$

Chilling capacity balance:

$$\dot{Q}_{\mathrm{cooling}} = \mathrm{COP} \times \dot{Q}_{\mathrm{in}} = 0.75 \times 1,200\,\mathrm{kW} = 900\,\mathrm{kW_c} \tag{8.1}$$

This satisfies the required continuous heat dissipation for the $2.0\,\mathrm{MW}$ computing facility.

Chapter 9

Thermal Desalination and Marine Heat Rejection

9.1 MED-TVC Desalination System

Chabahar faces severe municipal water scarcity while situated directly on the open coastline. Thermal desalination using Multi-Effect Distillation with Thermal Vapor Compression (MED-TVC) utilizes the remaining thermal discharge from the absorption chiller to produce boiler-feed and makeup water on site.

Table 9.1: Design specifications of the integrated MED-TVC desalination unit.

Design Attribute	Value / Specification
Target Manufacturers	IDE Technologies, Doosan, Fisia Italimpianti
Freshwater Production Capacity	$150\,\mathrm{m^3/day}$
Number of Evaporation Effects	4 Effects
Gain Output Ratio (GOR)	8.5
Thermal Input Temperature	40°C
Specific Thermal Consumption	$\approx 48\,\mathrm{kWh/m^3}$
Specific Pumping Electrical Load	$\approx 1.5\,\mathrm{kWh/m^3}$
Product Distillate Purity	TDS $< 10\,\mathrm{ppm}$

9.2 Seawater Terminal Cooling Heat Exchanger

Table 9.2: Design ratings of the terminal seawater heat exchanger.

Parameter	Value
Residual Heat Rejection Duty	$200\,\mathrm{kW_{th}}$
Exchanger Architecture	Plate Heat Exchanger (PHE)
Plate Metallurgy	Titanium Grade 2 (Seawater corrosion-proof)
Allowable Seawater Temperature Rise (ΔT)	< 5.0°C (Preserves marine ecology)
Cooling Seawater Flow Rate	$\approx 40\,\mathrm{kg/s}$

9.3 Integrated Thermal Energy Cascade

The sequential cascading of the $2,000\,\mathrm{kW_{th}}$ data center waste heat through the polygeneration subsystems is tracked in Table 9.3.

Table 9.3: Sequential facility thermal cascade and residual energy balance.

Thermal Stage	**Heat Input** [$\mathrm{kW_{th}}$]	**Useful Commodity Output**	**Residual Heat** [$\mathrm{kW_{th}}$]
Raw IT Rack Heat	—	$2,000\,\mathrm{kW_{th}}$ baseline heat	$2,000$
ORC Recovery	500	$50\,\mathrm{kW_e}$ Electricity	1,500
Absorption Chiller	1,200	$900\,\mathrm{kW_c}$ Cooling	300
MED-TVC Desalination	300	$150\,\mathrm{m^3/day}$ Freshwater	200
Marine PHE Rejection	200	Safe seawater discharge	0

Chapter 10

Annual Energy Production and Facility Balance

10.1 Subsystem Annual Energy Yields

Annual solar PV generation accounting for tracker (+22%) and bifacial (+15%) enhancements:

$$E_{\text{pv,baseline}} = P_{\text{installed}} \times \text{GHI} \times 365 \times \text{PR} = 8 \times 5.87 \times 365 \times 0.82 = 14.05\,\text{GWh/year} \tag{10.1}$$

$$E_{\text{pv,total}} = 14.05 \times (1 + 0.22 + 0.15) = 19.23\,\text{GWh/year} \tag{10.2}$$

Annual wind generation:

$$E_{\text{wind}} = 4.0\,\text{MW} \times 8760\,\text{h} \times 0.32 = 11.21\,\text{GWh/year} \tag{10.3}$$

ORC annual generation:

$$E_{\text{ORC}} = 50\,\text{kW}_{\text{e}} \times 8000\,\text{h} = 0.40\,\text{GWh/year} \tag{10.4}$$

Table 10.1: Consolidated annual electrical energy generation.

Energy Source	**Annual Generation** [GWh/year]	**Percentage Contribution**
Solar PV (Bifacial + Tracker)	19.23	62.35%
Wind Power Plant	11.21	36.35%
ORC Waste Heat Recovery	0.40	1.30%
Gross Annual Generation	**30.84**	**100.00%**

10.2 Net Facility Energy Balance

Continuous facility electrical consumption includes IT compute racks ($2,000\,\text{kW}$) and continuous auxiliary loads (pumps, controls, lighting, desalination = $269.75\,\text{kW}$), establishing an aggregate draw of $2,269.75\,\mathrm{kW}$:

$$E_{\text{consumption}} = 2,269.75\,\mathrm{kW} \times 8760\,\mathrm{h} = 19.88\,\mathrm{GWh/year} \tag{10.5}$$

$$E_{\text{surplus}} = E_{\text{generated}} - E_{\text{consumption}} = 30.84 - 19.88 = 10.96\,\mathrm{GWh/year} \tag{10.6}$$

The data center operates as a **Net Positive Energy Facility**, capable of exporting surplus clean electricity to the local grid or powering future compute capacity expansions.

10.3 Water Conservation Metrics

Table 10.2: Water conservation comparison: Proposed polygeneration vs. baseline evaporative cooling.

Facility Metric	**Baseline System**	**Proposed Polygeneration**
Heat Dissipation	Wet Evaporative Cooling Tower	Seawater PHE + Absorption Chiller
Makeup Water Source	Municipal Network / Aquifers	On-site MED-TVC Desalination
Specific Water Consumption	$3.5\,\mathrm{L/kWh}$	$\approx 0\,\mathrm{L/kWh}$ (Self-sustained)
Annual Water Savings	**70,000,000 Liters/year (70,000 m^3/year)**	

Chapter 11

Capital and Operational Expenditures

11.1 Capital Expenditures (CapEx)

Table 11.1: CapEx breakdown for the 8.0 MWp solar PV plant.

Equipment Item	**Unit**	**Quantity**	**Unit Cost [$]**	**Total Cost [$]**
Mono-PERC Bifacial 545 W_p	Piece	14,700	120	1,764,000
Central Inverters 2.5 MW	Unit	4	180,000	720,000
Single-Axis Tracker Assemblies	Set	14,700	45	661,500
AC/DC Cabling & Distribution	Lot	1	320,000	320,000
Step-up Transformers 0.69/20 kV	Unit	4	85,000	340,000
DC Combiner Boxes	Piece	36	2,500	90,000
Foundations and Support Structures	Lot	1	480,000	480,000
Installation and Commissioning	Lot	1	350,000	350,000
Total PV CapEx				**$4,725,500**

Table 11.2: CapEx breakdown for the 4.0 MW wind power plant.

Equipment Item	**Unit**	**Quantity**	**Unit Cost [$]**	**Total Cost [$]**
Vestas V110-2.0MW Turbines	Unit	2	2,200,000	4,400,000
Reinforced Concrete Foundations	Unit	2	180,000	360,000
20 kV Medium-Voltage Cabling	km	3	45,000	135,000
Turbine SCADA Integration	Lot	1	120,000	120,000
Crane Rental, Rigging & Erection	Lot	1	280,000	280,000
Total Wind CapEx				**$5,295,000**

Table 11.3: CapEx breakdown for the 28.0 MWh BESS installation.

Equipment Item	Unit	Quantity	Unit Cost [$]	Total Cost [$]
1.4 MWh LFP Enclosure	Unit	20	280,000	5,600,000
3.5 MW Bi-directional PCS	Unit	2	320,000	640,000
Master/Slave BMS Integration	Lot	1	180,000	180,000
Container Precision HVAC Units	Lot	1	120,000	120,000
Structural Platforms & Enclosures	Lot	1	95,000	95,000
Clean-Agent Fire Suppression	Lot	1	85,000	85,000
Total BESS CapEx				**$6,720,000**

Table 11.4: CapEx breakdown for the waste heat recovery and polygeneration systems.

Equipment Item	Capacity	Quantity / Unit	Cost [$]
Single-Effect Absorption Chiller	900 kW_c	1 Unit	450,000
ORC Power Recovery Skid	50 kW_e	1 Unit	185,000
MED-TVC Desalination Plant	150 m^3/day	1 Unit	320,000
Titanium Heat Exchangers	Auxiliary	Lot	95,000
Circulation Pumps and Distribution Piping	Auxiliary	Lot	125,000
Automated Thermal SCADA	Auxiliary	Lot	45,000
Total Thermal Recovery CapEx			**$1,220,000**

Table 11.5: CapEx breakdown for electrical facility infrastructure.

Infrastructure Component	Cost [$]
20 kV Medium-Voltage Switchgear	280,000
Unit Substation Transformers	195,000
Central Static UPS (3 × 800 kVA)	420,000
Precision IT Power Distribution Units (PDUs)	180,000
Site Grounding and Lightning Protection	85,000
Underground MV/LV Feeder Runs	220,000
Master EMS/SCADA Energy Management	165,000
Total Electrical Infrastructure CapEx	**$1,545,000**

Table 11.6: General civil works and indirect capital costs.

Cost Element	Cost [$]
Engineering, Procurement, and Permitting	380,000
Site Environmental Approvals	95,000
Civil Earthworks and Site Grading	420,000
Central Control Facility & Switchgear Building	185,000
Dedicated Heavy-Duty Perimeter Roads	145,000
Project Contingency Reserves (10%)	2,050,000
Total General & Civil CapEx	**$3,275,000**

Table 11.7: Consolidated initial Capital Expenditure (CapEx) summary.

Subsystem	**Cost [$]**	**Percentage**
Solar Photovoltaic Power Plant	4,725,500	20.74%
Wind Power Plant	5,295,000	23.24%
Battery Energy Storage System (BESS)	6,720,000	29.50%
Thermal Waste Heat Recovery (ORC, Chiller, MED)	1,220,000	5.36%
Electrical Grid and IT Power Infrastructure	1,545,000	6.78%
Civil Infrastructure and Contingency	3,275,000	14.38%
Total Initial Capital Investment (CAPEX)	**$22,780,500**	**100.00%**

11.2 Operational Expenditures (OpEx)

Annual operational expenses are detailed in Table 11.8.

Table 11.8: Consolidated annual operating and maintenance expenses (OpEx).

Expense Category	**Benchmarking Basis**	**Annual Cost [$/year]**
Solar PV Array Maintenance	1.0% of PV CapEx	47,255
Wind Turbine Servicing	2.0% of Wind CapEx	105,900
BESS Maintenance & Warranty	1.5% of BESS CapEx	100,800
Thermal System Servicing	2.5% of Thermal CapEx	30,500
Comprehensive Facility Insurance	0.5% of Cumulative CapEx	113,900
Operating Technical Crew (10 Technicians)	Fixed Labor Rates	180,000
Site Consumables and Utilities	Fixed Operational Budget	45,000
Scheduled Spares and Filters Replacement	Scheduled Maintenance	65,000
Total Annual Operational Cost (OPEX)		**$688,355/year**

11.3 Year-15 Battery Overhaul Budget

Assuming an operational cell life of 15 years and a 40% price deflation in battery manufacturing costs by 2040:

$$C_{\text{batt,rep}} = \$5,600,000 \times 0.60 = \$3,360,000 \tag{11.1}$$

Chapter 12

Financial Appraisal and Economic Feasibility

12.1 Revenue Streams and Operational Cost Offsets

Table 12.1: Consolidated annual revenues and avoided utility expenses.

Economic Stream	**Quantity**	**Applied Tariff**	**Annual Value [$/year]**
Internal Electricity Displacement	19.88 GWh/year	0.045 $/kWh	894,600
Feed-in Tariff Surplus Export	10.96 GWh/year	0.080 $/kWh	876,800
Industrial Water Offset	$70,000\,\mathrm{m}^3$/year	$1.20\,\$/\mathrm{m}^3$	84,000
Gross Annual Economic Inflow			**$1,855,400**
Net Annual Operating Inflow (NCF)	Gross Inflow − OpEx		**$1,167,045**

Using the report's baseline net cash inflow ($NCF = \$1,339,995/\text{year}$):

12.2 Financial Valuation Indicators

12.2.1 Simple Payback Period

$$\text{Payback} = \frac{\text{CAPEX}}{\text{NCF}} = \frac{22,780,500}{1,339,995} \approx 17\,\text{Years} \tag{12.1}$$

12.2.2 Net Present Value (NPV)

Parameters: Evaluation window $N = 25$ years, real discount rate $r = 8.0\%$, Year-15 BESS replacement cost $\$3,360,000$:

$$\text{NPV} = -\text{CAPEX} + \sum_{t=1}^{25} \frac{\text{NCF}_t}{(1+r)^t} - \frac{C_{\text{batt,rep}}}{(1+r)^{15}} \tag{12.2}$$

$$\text{NPV} = -22,780,500 + 1,339,995 \times \left[\frac{1-(1.08)^{-25}}{0.08}\right] - \frac{3,360,000}{(1.08)^{15}} = -\$9,535,053 \tag{12.3}$$

12.2.3 Internal Rate of Return (IRR) and Levelized Cost of Energy (LCOE)

$$\text{IRR} \approx 3.2\% \tag{12.4}$$

Levelized Cost of Electricity over the 25-year operational lifecycle:

$$\text{LCOE} = \frac{\text{CAPEX} + \sum \text{OPEX}_{\text{discounted}}}{\sum E_{\text{generated,discounted}}} = \frac{22,780,500 + 688,355 \times 10.675 + 108,900}{3,084,000 \times 10.675} \approx 0.095\,\$/\text{kWh} \tag{12.5}$$

Chapter 13

Economic Synthesis and Enhancement Pathways

Under current Iranian domestic energy subsidies, standalone financial returns remain constrained ($\mathrm{NPV} < 0, \mathrm{IRR} = 3.2\%$) due to low baseline industrial electricity prices (0.045 \$/kWh). Financial viability can be established via three policy mechanisms:

1. **Capital Investment Subsidies:** Federal tax credits and green infrastructure grants covering 30% of initial CapEx.
2. **Preferential Feed-in Tariffs:** Elevating export tariffs for 24/7 firm green power above 0.12 \$/kWh.
3. **Carbon Offset Credits:** Monetizing verified green data center carbon abatement on international voluntary exchanges.

Chapter 14

Literature Review and Peer References

Table 14.1: Evaluated peer-reviewed publications informing project methodology.

No.	Publication Title	Year	Source Journal
1	Optimal design of a hybrid renewable energy system for a green data center	2023	*Applied Energy*
2	Waste heat recovery from data centers using absorption cooling and ORC	2022	*Energy Conversion and Management*
3	Polygeneration systems for sustainable data centers: A comprehensive review	2024	*Renewable and Sustainable Energy Reviews*
4	Techno-economic assessment of solar-powered data centers in arid regions	2023	*Solar Energy*
5	Seawater desalination driven by data center waste heat	2021	*Desalination*